EXPERIMENT-3

3. Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.

Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

Aim

To find the average sale price of houses having more than four bedrooms using NumPy arrays.

Algorithm

1. Import the NumPy library.
2. Create a NumPy array containing bedrooms, square footage, and sale price.
3. Select houses having more than four bedrooms.
4. Extract the sale prices of the selected houses.
5. Calculate their average using np.mean().
6. Display the average sale price.

Code

```
import numpy as np


# Columns: Bedrooms, Square Feet, Sale Price
house_data = np.array([
    [3, 1200, 250000],
```

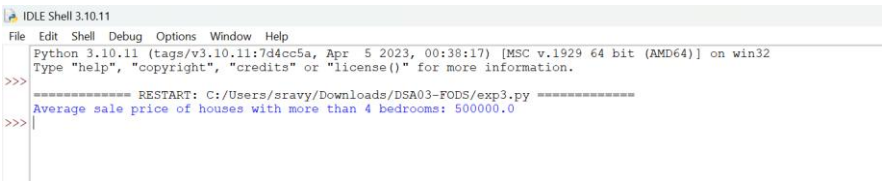
```
    [5, 2000, 450000],
    [4, 1800, 350000],
    [6, 2500, 550000],
    [5, 2200, 500000]
])
selected_houses = house_data[house_data[:, 0] > 4]
average_price = np.mean(selected_houses[:, 2])
print("Average sale price of houses with more than 4 bedrooms:",
      average_price)
```

Input

```
3  1200  250000
5  2000  450000
4  1800  350000
6  2500  550000
5  2200  500000
```

Output

Average sale price of houses with more than 4 bedrooms: 500000.0



Result

Thus, the average sale price of houses with more than four bedrooms was calculated successfully as **500000**.